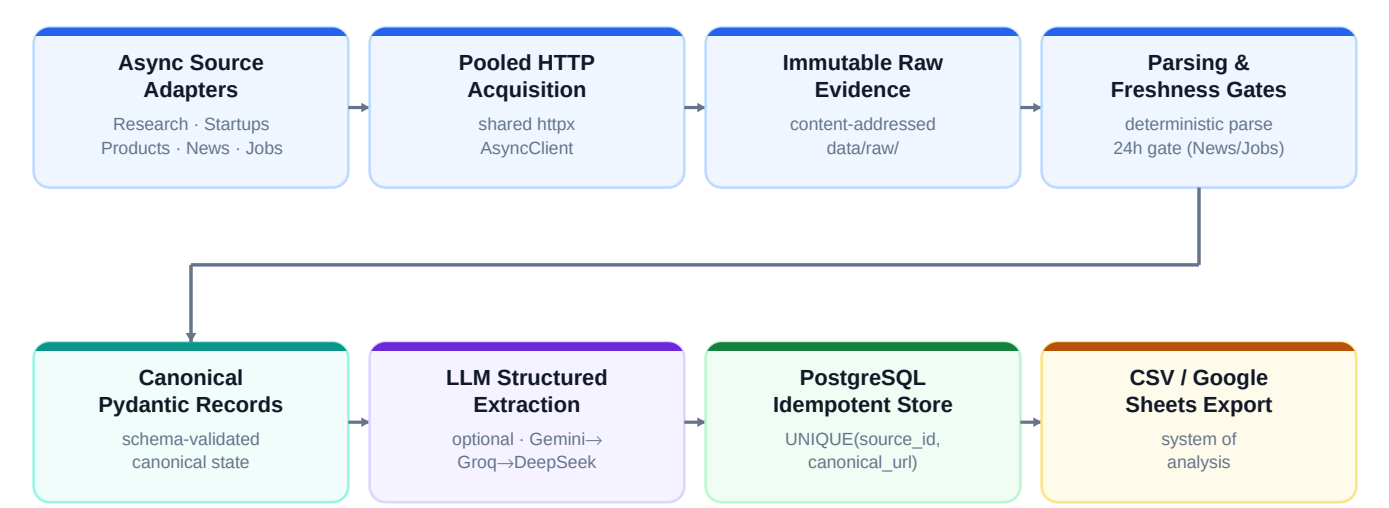


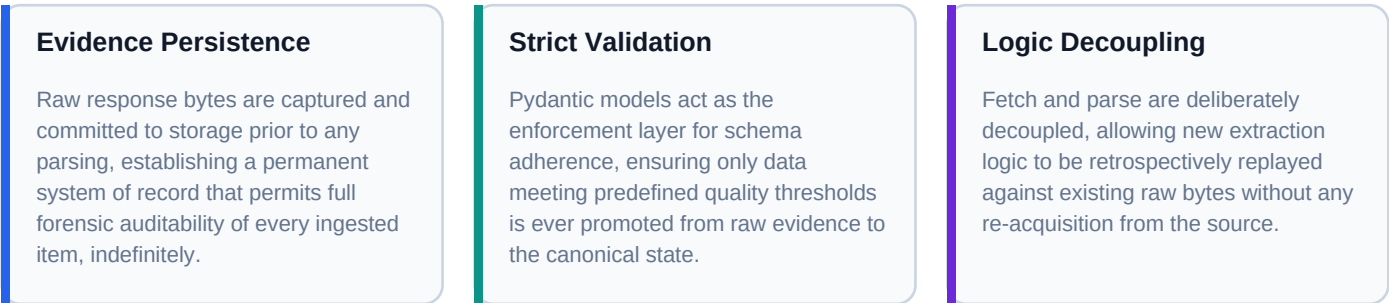
1 High-Level Architecture

A **provenance-first** pipeline: no source fact is inferred without evidence. Acquisition is strategically decoupled from extraction, so raw evidence is captured once, permanently, and extraction logic can be re-applied later against the original bytes without ever re-fetching from the source.

END-TO-END PIPELINE FLOW



CORE DESIGN PRINCIPLES



CROSS-CUTTING CONCERNS

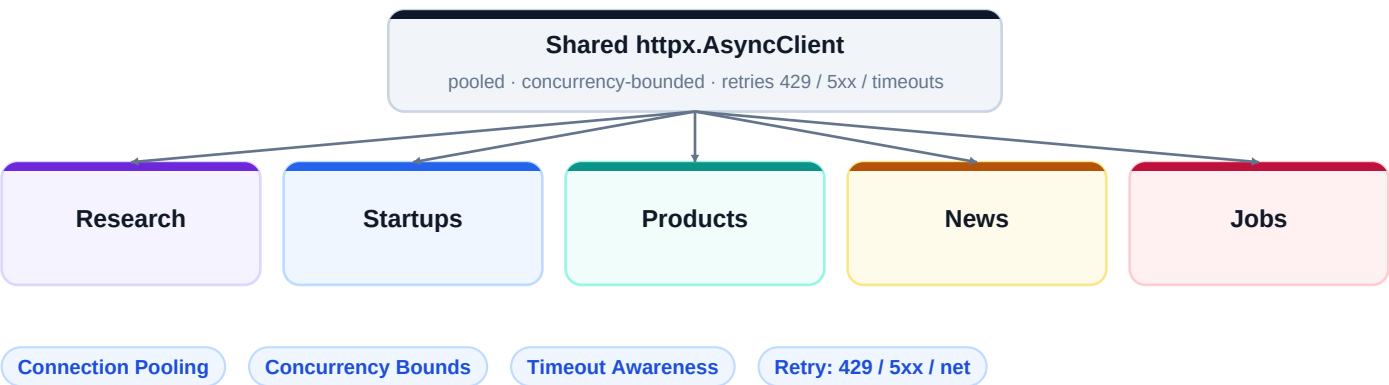


MULTI-TIER PROCESSING SEQUENCE
async source adapters → pooled HTTP acquisition → immutable raw evidence → deterministic parsing & freshness gates → canonical Pydantic records → optional LLM structured extraction → PostgreSQL idempotent persistence → CSV / Google Sheets export

2 Acquisition, Adaptation & Data Integrity

2 · MULTI-VERTICAL ACQUISITION & CONCURRENCY

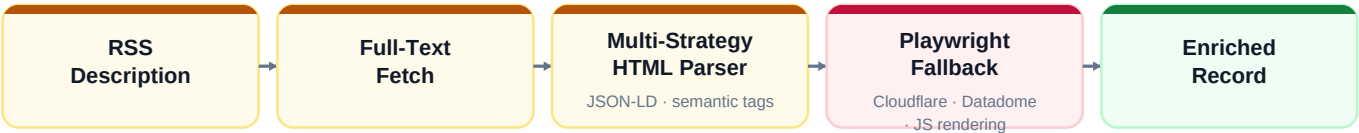
All outgoing traffic is aggregated through one pooled HTTP engine, enforcing global rate limits and connection reuse across five source verticals — protecting the pipeline's standing with every provider.



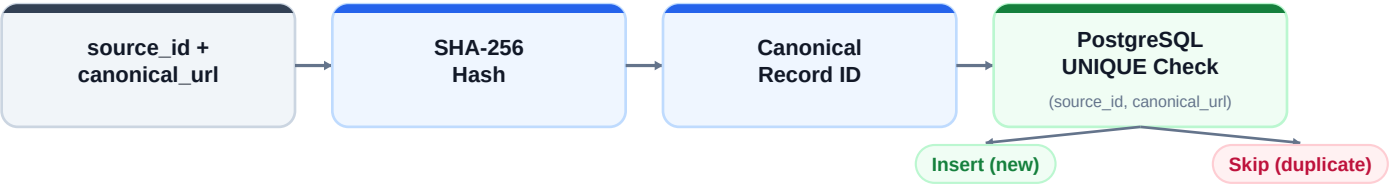
3 · SOURCE-SPECIFIC ADAPTATION & ENRICHMENT

Vertical	Sources	Target Volume / Gate
Research	arXiv (paginated API) · Papers with Code (official implementations only)	1,000 records each
Startups	Y Combinator via Algolia API	1,000 companies
Products	Product Hunt · There's An AI For That	1,000 products each
News	TechCrunch · VentureBeat · MIT Tech Review · OpenAI · Google AI	24-hour freshness gate
Jobs	Remote OK · We Work Remotely · Himalayas · Jobspresso · AI-Jobs.net	24-hour freshness gate

News / Jobs enrichment — full-text is used only when longer than the RSS description:



4 · DETERMINISTIC IDENTITY & IDEMPOTENT PERSISTENCE

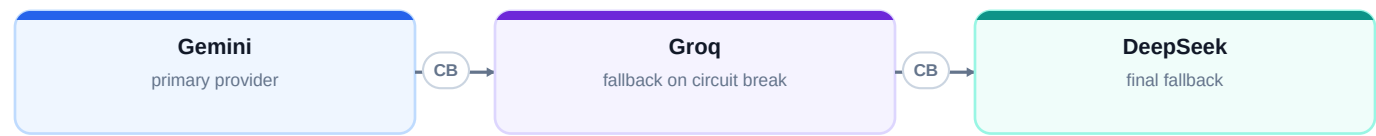


Raw bytes are written to content-addressed files in **data/raw/** before parsing — enabling forensic replay against immutable evidence without ever re-scraping the source. External enrichment such as GitHub-star counts is cached by normalized repository URL, so concurrent paper-to-repository lookups within a run share a single API call. Because PostgreSQL enforces **UNIQUE(source_id, canonical_url)**, retrying the same item can never create a second canonical record.

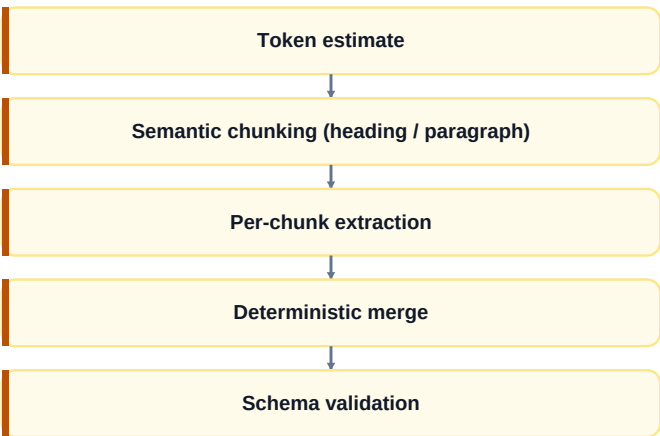
3 LLM Extraction, Recovery & Scalability

5 · MULTI-TIER LLM EXTRACTION & RECOVERY LAYER

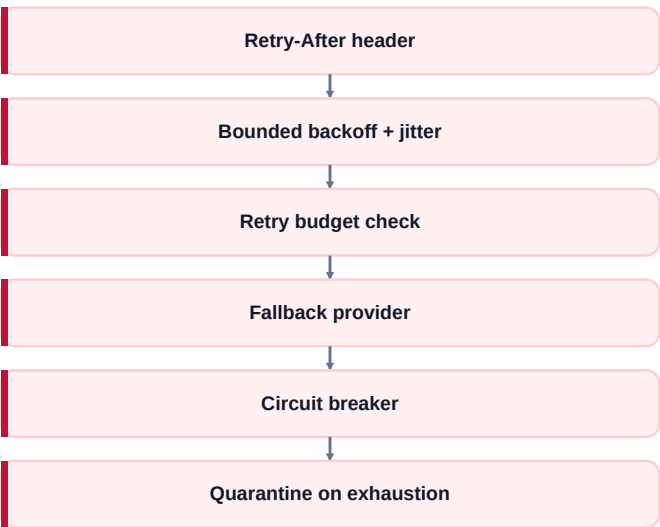
Unstructured text is transformed into structured intelligence through a provider fallback chain, with circuit breakers, adaptive oversized-payload recovery, and strict schema validation on every response.



413 · PAYLOAD TOO LARGE



429 · RATE LIMIT



6 · SCALABILITY & EVOLUTION ROADMAP

Small-Boundary Storage Migration



System of Record → System of Analysis



Horizontal scale: 500k+ records flow through a partitioned queue to stateless workers backed by shared durable storage — capacity grows simply by adding workers, bounded by source/provider limits and backpressure.

Conclusion

A robust, future-proof foundation for high-fidelity intelligence gathering.

- Provenance-First
- Idempotent Persistence
- Multi-Tier LLM Redundancy
- Analyst-Ready Export

Source: github.com/vineet-777/AI-Internship-DemoTask